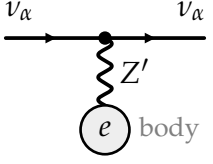
Mediator X: Z' ($L_e - L_\alpha$), Z_D (dark portal), V or ϕ (secret interactions; ϕ's own diagrams in the appendix)

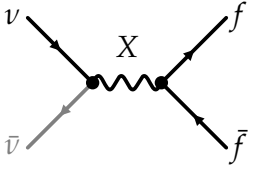
Beam neutrinos black, relic neutrinos gray; f: a neutrino or a charged lepton

(a) Potential of matter



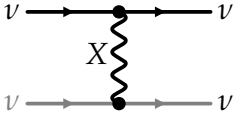
$L_e - L_\alpha$

(b) Absorption, s channel



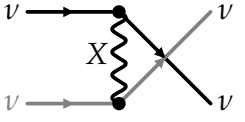
All models

(c) Scattering, t channel



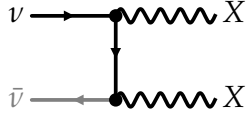
All models

(d) Scattering, u channel



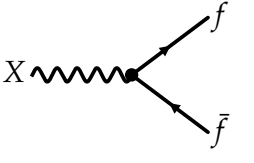
All models

(e) Pair production



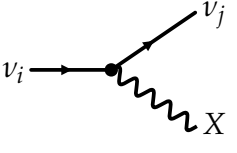
All models

(f) Mediator decay



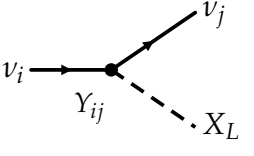
All models

(g) Neutrino decay



$L_e - L_\alpha$, dark portal

(h) Longitudinal mode



Vector mediators